

L^AT_EX Cheat Sheet

Document classes

<code>book</code>	Default is two-sided.
<code>report</code>	No <code>\part</code> divisions.
<code>article</code>	No <code>\part</code> or <code>\chapter</code> divisions.
<code>letter</code>	Letter (?).
<code>slides</code>	Large sans-serif font.

Used at the very beginning of a document: `\documentclass{class}`.
Use `\begin{document}` to start contents and `\end{document}` to end the document.

Common documentclass options

<code>10pt/11pt/12pt</code>	Font size.
<code>letterpaper/a4paper</code>	Paper size.
<code>twocolumn</code>	Use two columns.
<code>twoside</code>	Set margins for two-sided.
<code>landscape</code>	Landscape orientation. Must use <code>dvips -t landscape</code> .
<code>draft</code>	Double-space lines.
Usage: <code>\documentclass[opt,opt]{class}</code> .	

Packages

<code>fullpage</code>	Use 1 inch margins.
<code>anysize</code>	Set margins: <code>\marginsize{l}{r}{t}{b}</code> .
<code>multicol</code>	Use n columns: <code>\begin{multicols}{n}</code> .
<code>latexsym</code>	Use L ^A T _E X symbol font.
<code>graphicx</code>	Show image: <code>\includegraphics[width=x]{file}</code> .
<code>url</code>	Insert URL: <code>\url{http://...}</code> .
Use before <code>\begin{document}</code> . Usage: <code>\usepackage{package}</code>	

Title

<code>\author{text}</code>	Author of document.
<code>\title{text}</code>	Title of document.
<code>\date{text}</code>	Date.
These commands go before <code>\begin{document}</code> . The declaration <code>\maketitle</code> goes at the top of the document.	

Miscellaneous

<code>\pagestyle{empty}</code>	Empty header, footer and no page numbers.
<code>\tableofcontents</code>	Add a table of contents here.

Document structure

<code>\part{title}</code>	<code>\subsubsection{title}</code>
<code>\chapter{title}</code>	<code>\paragraph{title}</code>
<code>\section{title}</code>	<code>\subparagraph{title}</code>
<code>\subsection{title}</code>	
Use <code>\setcounter{secnumdepth}{x}</code> suppresses heading numbers of depth $> x$, where <code>chapter</code> has depth 0. Use a <code>*</code> , as in <code>\section*{title}</code> , to not number a particular item—these items will also not appear in the table of contents.	

Text environments

<code>\begin{comment}</code>	Comment (not printed). Requires <code>verbatim</code> package.
<code>\begin{quote}</code>	Indented quotation block.
<code>\begin{quotation}</code>	Like <code>quote</code> with indented paragraphs.
<code>\begin{verse}</code>	Quotation block for verse.

Lists

<code>\begin{enumerate}</code>	Numbered list.
<code>\begin{itemize}</code>	Bulleted list.
<code>\begin{description}</code>	Description list.
<code>\item text</code>	Add an item.
<code>\item[x] text</code>	Use x instead of normal bullet or number. Required for descriptions.

References

<code>\label{marker}</code>	Set a marker for cross-reference, often of the form <code>\label{sec:item}</code> .
<code>\ref{marker}</code>	Give section/body number of marker.
<code>\pageref{marker}</code>	Give page number of marker.
<code>\footnote{text}</code>	Print footnote at bottom of page.

Floating bodies

<code>\begin{table}[place]</code>	Add numbered table.
<code>\begin{figure}[place]</code>	Add numbered figure.
<code>\begin{equation}[place]</code>	Add numbered equation.
<code>\caption{text}</code>	Caption for the body.
The <i>place</i> is a list valid placements for the body. <code>t</code> =top, <code>h</code> =here, <code>b</code> =bottom, <code>p</code> =separate page, <code>!</code> =place even if ugly. Captions and label markers should be within the environment.	

Text properties

Font face

Command	Declaration	Effect
<code>\textrm{text}</code>	<code>\rmfamily text</code>	Roman family
<code>\textsf{text}</code>	<code>\sffamily text</code>	Sans serif family
<code>\texttt{text}</code>	<code>\ttfamily text</code>	Typewriter family
<code>\textmd{text}</code>	<code>\mdseries text</code>	Medium series
<code>\textbf{text}</code>	<code>\bfseries text</code>	Bold series
<code>\textup{text}</code>	<code>\upshape text</code>	Upright shape
<code>\textit{text}</code>	<code>\itshape text</code>	<i>Italic shape</i>
<code>\textsl{text}</code>	<code>\slshape text</code>	<i>Slanted shape</i>
<code>\textsc{text}</code>	<code>\scshape text</code>	SMALL CAPS SHAPE
<code>\emph{text}</code>	<code>\em text</code>	<i>Emphasized</i>
<code>\textnormal{text}</code>	<code>\normalfont text</code>	Document font
<code>\underline{text}</code>		<u>Underline</u>

The command `(ttt)` form handles spacing better than the declaration `(ttt)` form.

Font size

<code>\tiny</code>	tiny	<code>\Large</code>	Large
<code>\scriptsize</code>	scriptsize	<code>\LARGE</code>	LARGE
<code>\footnotesize</code>	footnotesize		
<code>\small</code>	small	<code>\huge</code>	huge
<code>\normalsize</code>	normalsize		
<code>\large</code>	large	<code>\Huge</code>	Huge

These are declarations and should be used in the form `\small ...`, or without braces to affect the entire document.

Verbatim text

<code>\begin{verbatim}</code>	Verbatim environment.
<code>\begin{verbatim*}</code>	Spaces are shown as <code>␣</code> .
<code>\verb!text!</code>	Text between the delimiting characters (in this case ‘!’) is verbatim.

Justification

Environment	Declaration
<code>\begin{center}</code>	<code>\centering</code>
<code>\begin{flushleft}</code>	<code>\raggedright</code>
<code>\begin{flushright}</code>	<code>\raggedleft</code>

Miscellaneous

`\linespread{x}` changes the line spacing by the multiplier x .

Text-mode symbols

Symbols

<code>&</code>	<code>\&</code>	<code>-</code>	<code>_</code>	<code>...</code>	<code>\ldots</code>	<code>•</code>	<code>\textbullet</code>
<code>\$</code>	<code>\\$</code>	<code>^</code>	<code>\^{}{}</code>	<code> </code>	<code>\textbar</code>	<code>\</code>	<code>\textbackslash</code>
<code>%</code>	<code>\%</code>	<code>~</code>	<code>\~{}{}</code>	<code>#</code>	<code>\#</code>	<code>§</code>	<code>\S</code>

Accents

<code>ò</code>	<code>\'o</code>	<code>ó</code>	<code>\'o</code>	<code>ô</code>	<code>\^o</code>	<code>õ</code>	<code>\~o</code>	<code>ö</code>	<code>=o</code>
<code>ó</code>	<code>\.o</code>	<code>ö</code>	<code>\"o</code>	<code>q</code>	<code>\c o</code>	<code>ô</code>	<code>\v o</code>	<code>ö</code>	<code>\H o</code>
<code>ç</code>	<code>\c c</code>	<code>q</code>	<code>\d o</code>	<code>q</code>	<code>\b o</code>	<code>ö</code>	<code>\t oo</code>	<code>œ</code>	<code>\oe</code>
<code>Œ</code>	<code>\OE</code>	<code>æ</code>	<code>\ae</code>	<code>Æ</code>	<code>\AE</code>	<code>â</code>	<code>\aa</code>	<code>Å</code>	<code>\AA</code>
<code>ø</code>	<code>\o</code>	<code>Ø</code>	<code>\O</code>	<code>ı</code>	<code>\l</code>	<code>L</code>	<code>\L</code>	<code>ı</code>	<code>\i</code>
<code>j</code>	<code>\j</code>	<code>i</code>	<code>\~{}</code>	<code>ı</code>	<code>\?</code>				

Delimiters

<code>'</code>	<code>\'</code>	<code>[</code>	<code>\[</code>	<code>(</code>	<code>\(</code>	<code><</code>	<code>\textless</code>
<code>,</code>	<code>\,</code>	<code>]</code>	<code>\]</code>	<code>)</code>	<code>\)</code>	<code>></code>	<code>\textgreater</code>

Dashes

Name	Source	Example	Usage
<code>hyphen</code>	<code>-</code>	X-ray	In words.
<code>en-dash</code>	<code>--</code>	1–5	Between numbers.
<code>em-dash</code>	<code>---</code>	Yes—or no?	Punctuation.

Line and page breaks

<code>\</code>	Begin new line without new paragraph.
<code>*</code>	Prohibit pagebreak after linebreak.
<code>\kill</code>	Don't print current line.
<code>\pagebreak</code>	Start new page.
<code>\noindent</code>	Do not indent current line.

Miscellaneous

<code>\today</code>	September 20, 2025.
<code>\$\sim\$</code>	Prints <code>~</code> instead of <code>\~{}</code> , which makes <code>~</code> .
<code>~</code>	Space, disallow linebreak (W.J.~Clinton).
<code>\@.</code>	Indicate that the <code>.</code> ends a sentence when following an uppercase letter.
<code>\hspace{l}</code>	Horizontal space of length l (Ex: $l = 20\text{pt}$).
<code>\vspace{l}</code>	Vertical space of length l .
<code>\rule{w}{h}</code>	Line of width w and height h .

Tabular environments

tabbing environment

<code>\=</code>	Set tab stop.	<code>\></code>	Go to tab stop.
Tab stops can be set on “invisible” lines with <code>\kill</code> at the end of the line. Normally <code>\</code> is used to separate lines.			

tabular environment

`\begin{array}[pos]{cols}`
`\begin{tabular}[pos]{cols}`
`\begin{tabular*}{width}[pos]{cols}`

tabular column specification

`l` Left-justified column.
`c` Centered column.
`r` Right-justified column.
`p{width}` Same as `\parbox[t]{width}`.
`@{decl}` Insert *decl* instead of inter-column space.
`|` Inserts a vertical line between columns.

tabular elements

`\hline` Horizontal line between rows.
`\cline{x-y}` Horizontal line across columns *x* through *y*.
`\multicolumn{n}{cols}{text}`
A cell that spans *n* columns, with *cols* column specification.

Math mode

For inline math, use `\(...\)` or `$...$`. For displayed math, use `\[...\]` or `\begin{equation}`.

Superscript^{*x*} `~{x}` Subscript_{*x*} `_ {x}`
 $\frac{x}{y}$ `\frac{x}{y}` $\sum_{k=1}^n$ `\sum_{k=1}^n`
 $\sqrt[n]{x}$ `\sqrt[n]{x}` $\prod_{k=1}^n$ `\prod_{k=1}^n`

Math-mode symbols

<code>\leq</code>	<code>\geq</code>	<code>\neq</code>	<code>\approx</code>
<code>\times</code>	<code>\div</code>	<code>\pm</code>	<code>\cdot</code>
<code>^\circ</code>	<code>\circ</code>	<code>\prime</code>	<code>\cdots</code>
<code>\infty</code>	<code>\neg</code>	<code>\wedge</code>	<code>\vee</code>
<code>\supset</code>	<code>\forall</code>	<code>\in</code>	<code>\rightarrow</code>
<code>\subset</code>	<code>\exists</code>	<code>\notin</code>	<code>\Rightarrow</code>
<code>\cup</code>	<code>\cap</code>	<code> </code>	<code>\Leftrightarrow</code>
<code>\dot{a}</code>	<code>\hat{a}</code>	<code>\bar{a}</code>	<code>\tilde{a}</code>
<code>\alpha</code>	<code>\beta</code>	<code>\gamma</code>	<code>\delta</code>
<code>\epsilon</code>	<code>\zeta</code>	<code>\eta</code>	<code>\varepsilon</code>
<code>\theta</code>	<code>\iota</code>	<code>\kappa</code>	<code>\vartheta</code>
<code>\lambda</code>	<code>\mu</code>	<code>\nu</code>	<code>\xi</code>
<code>\pi</code>	<code>\rho</code>	<code>\sigma</code>	<code>\tau</code>
<code>\upsilon</code>	<code>\phi</code>	<code>\chi</code>	<code>\psi</code>
<code>\omega</code>	<code>\Gamma</code>	<code>\Delta</code>	<code>\Theta</code>
<code>\Lambda</code>	<code>\Xi</code>	<code>\Pi</code>	<code>\Sigma</code>
<code>\Upsilon</code>	<code>\Phi</code>	<code>\Psi</code>	<code>\Omega</code>

SI Units

With `\usepackage{siunitx}` `\verb` you can use correctly formatted SI units. Simply write `\si{<unit name>}` for single units and to write quantities use `\SI[<option>]{<quantity>}{<units>}`, e.g., `8.85 \times 10^{-12} \frac{C^2 m^2}{N} \SI[per-mode = fraction]{8.85e-12}{\coulomb\squared\per\newton\meter\squared}`. However, it is important to note that the quantity needs to me a number, e.g., `8.85e-12` and not using commands such as `8.85\times 10^{-12}` where both `\times` and `10^{-12}` is problematic. There are multiple versions of *siunitx*, the syntax shown is for the old version.

Base SI Units

`m` `\si{meter}` or `\si{m}`
`kg` `\si{kilogram}` or `\si{kg}`
`s` `\si{second}` or `\si{s}`
`A` `\si{ampere}` or `\si{A}`
`K` `\si{kelvin}` or `\si{K}`
`mol` `\si{mole}` or `\si{mol}`
`cd` `\si{candela}` or `\si{cd}`

Derived SI Units

`Hz` `\si{hertz}` or `\si{Hz}`
`N` `\si{newton}` or `\si{N}`
`Pa` `\si{pascal}` or `\si{Pa}`
`J` `\si{joule}` or `\si{J}`
`W` `\si{watt}` or `\si{W}`
`C` `\si{coulomb}` or `\si{C}`
`V` `\si{volt}` or `\si{V}`
`Ω` `\si{ohm}`
`S` `\si{siemens}` or `\si{S}`
`Wb` `\si{weber}` or `\si{Wb}`
`T` `\si{tesla}` or `\si{T}`
`H` `\si{henry}` or `\si{H}`
`°C` `\si{degreeCelsius}`
`lm` `\si{lumen}` or `\si{lm}`
`lx` `\si{lux}` or `\si{lx}`
`Bq` `\si{becquerel}` or `\si{Bq}`
`Gy` `\si{gray}` or `\si{Gy}`
`kat` `\si{katal}` or `\si{kat}`
`rad` `\si{radian}`
`sr` `\si{steradian}`

SI Prefixes

10^{24}	<code>Y</code>	<code>\si{yotta}</code>	10^{21}	<code>Z</code>	<code>\si{zetta}</code>
10^{18}	<code>E</code>	<code>\si{exa}</code>	10^{15}	<code>P</code>	<code>\si{peta}</code>
10^{11}	<code>T</code>	<code>\si{tera}</code>	10^9	<code>G</code>	<code>\si{giga}</code>
10^6	<code>M</code>	<code>\si{mega}</code>	10^3	<code>k</code>	<code>\si{kilo}</code>
10^2	<code>h</code>	<code>\si{hecto}</code>	10^1	<code>da</code>	<code>\si{deka}</code>
10^{-1}	<code>d</code>	<code>\si{deci}</code>	10^{-2}	<code>c</code>	<code>\si{centi}</code>
10^{-3}	<code>m</code>	<code>\si{milli}</code>	10^{-6}	<code>μ</code>	<code>\si{micro}</code>
10^{-9}	<code>n</code>	<code>\si{nano}</code>	10^{-12}	<code>p</code>	<code>\si{pico}</code>
10^{-15}	<code>f</code>	<code>\si{femto}</code>	10^{-18}	<code>a</code>	<code>\si{atto}</code>
10^{-21}	<code>z</code>	<code>\si{zepto}</code>	10^{-24}	<code>y</code>	<code>\si{yocto}</code>

Bibliography and citations

When using BibTeX, you need to run `latex`, `bibtex`, and `latex` twice more to resolve dependencies.

Citation types

`\cite{key}` Full author list and year. (Watson and Crick 1953)
`\citeA{key}` Full author list. (Watson and Crick)
`\citeN{key}` Full author list and year. Watson and Crick (1953)
`\shortcite{key}` Abbreviated author list and year. ?
`\shortciteA{key}` Abbreviated author list. ?
`\shortciteN{key}` Abbreviated author list and year. ?
`\citeyear{key}` Cite year only. (1953)
All the above have an NP variant without parentheses; Ex. `\citeNP`.

BibTeX entry types

`@article` Journal or magazine article.
`@book` Book with publisher.
`@booklet` Book without publisher.
`@conference` Article in conference proceedings.
`@inbook` A part of a book and/or range of pages.
`@incollection` A part of book with its own title.
`@misc` If nothing else fits.
`@phdthesis` PhD. thesis.
`@proceedings` Proceedings of a conference.
`@techreport` Tech report, usually numbered in series.
`@unpublished` Unpublished.

BibTeX fields

`address` Address of publisher. Not necessary for major publishers.
`author` Names of authors, of format
`booktitle` Title of book when part of it is cited.
`chapter` Chapter or section number.
`edition` Edition of a book.
`editor` Names of editors.
`institution` Sponsoring institution of tech. report.
`journal` Journal name.
`key` Used for cross ref. when no author.
`month` Month published. Use 3-letter abbreviation.
`note` Any additional information.
`number` Number of journal or magazine.
`organization` Organization that sponsors a conference.
`pages` Page range (2,6,9--12).
`publisher` Publisher's name.
`school` Name of school (for thesis).
`series` Name of series of books.
`title` Title of work.
`type` Type of tech. report, ex. "Research Note".
`volume` Volume of a journal or book.
`year` Year of publication.
Not all fields need to be filled. See example below.

Common BibTeX style files

<code>abbrv</code>	Standard	<code>abstract</code>	alpha with abstract
<code>alpha</code>	Standard	<code>apa</code>	APA
<code>plain</code>	Standard	<code>unsrt</code>	Unsorted

The L^AT_EX document should have the following two lines just before `\end{document}`, where `bibfile.bib` is the name of the BibTeX file.

```
\bibliographystyle{plain}
\bibliography{bibfile}
```

BibTeX example

The BibTeX database goes in a file called `file.bib`, which is processed with `bibtex` file.

```
@String{N = {Na\-ture}}
@Article{WC:1953,
  author = {James Watson and Francis Crick},
  title = {A structure for Deoxyribose Nucleic Acid},
  journal = N,
```

```
volume = {171},
pages  = {737},
year   = 1953
}
```

Sample L^AT_EX document

```
\documentclass[11pt]{article}
\usepackage{fullpage}
\title{Template}
\author{Name}
\begin{document}
\maketitle
```

```
\section{section}
\subsection*{subsection without number}
text \textbf{bold text} text. Some math:  $2+2=5$ 
\subsection{subsection}
text \emph{emphasized text} text. \cite{WC:1953}
discovered the structure of DNA.
```

```
A table:
\begin{table}[!th]
\begin{tabular}{|l|c|r|}
\hline
first & row & data \\
```

```
second & row & data \\
\hline
\end{tabular}
\caption{This is the caption}
\label{ex:table}
\end{table}
```

```
The table is numbered \ref{ex:table}.
\end{document}
```

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<http://wch.github.io/latexsheet/>